

1)

a)

Gilberts model can be used to give a picture of the magnetic dipole in similar terms as the electric dipole when looking at the formula for magnetic torque, as in this model they have more resemblances.

b)

When looking at the way the magnetic material behaves for each dipole, then the Ampere's model is more correct as this accounts for that the dipoles come from current loops and not a pair of magnetic charges.

c)

It is similar in terms of having two opposite charges or magnetic poles separated by a distance. But when looking at the dipoles, they have different reasons for coming to be, being charges and current loops.

2)

a)

Magnetic susceptibility is a dimensionless quantity, varying by different materials, that represents the constant of proportionality between the magnetic dipole moment M and the auxiliary field H .

$$M = \chi_m H$$

For a linear media, we have that:

$$B = \mu_0(H + M) = \mu_0(1 + \chi_m)H = \mu H$$

where

$$\mu = \mu_0(1 + \chi_m).$$

b)

The auxiliary field H can be found using Ampere's law for the free currents running through the solenoid. But the magnetic field B is dependent on the bound currents as well, and therefore often is calculated later in terms of the magnetic dipole moment and auxiliary fields.

c)

The magnetic field inside the solenoid changes whether it's a paramagnetic or a diamagnetic due to the formula $B = \mu_0(1 + \chi_m)H$ where the exponent of χ_m is dependent on the material (positive for paramagnetic, negative for diamagnetic). Thus the magnetic field will be stronger for paramagnetic material and weaker for the diamagnetic. The physical explanation for this is the induced magnetic dipole moment will go with or against the magnetic field depending on the material, either strengthening or weakening the magnetic field.

d)

- I. $\chi_m > 0$ gives a magnetic dipole moment with the B-field ($+\phi$) and thus K_b will point down from the right hand rule of cross product.
- II. $\chi_m < 0$ gives a magnetic dipole moment against the B-field ($+\phi$) and thus K_b will point up from the right hand rule of cross product.

e)

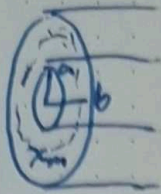
The equation:

$$J_b = \nabla \times M = \nabla \times (\chi_m H) = \chi_m J_f$$

This means that the bound currents are proportional to the free currents through the material. This comes from the Ampere's law for the auxiliary field, and the proportionality between B and H. When the free currents run through the material, then the bound currents will only exist on the surface of the material.

3)

3) a) $a < s < b$



$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}$$

$$\Rightarrow 2\pi s |\mathbf{H}| = I$$

$$\Rightarrow \mathbf{H} = \frac{I}{2\pi s} \hat{\varphi}$$

$$\mathbf{B} = \mu \mathbf{H} = \mu_0(1 + \chi_m) \mathbf{H} = \frac{\mu_0 I (1 + \chi_m)}{2\pi s} \hat{\varphi}$$

$$b) \mathbf{M} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{H} = \frac{I}{2\pi s} (-1 + (1 + \chi_m)) \hat{\varphi} = \frac{-I}{2\pi s} \chi_m \hat{\varphi}$$

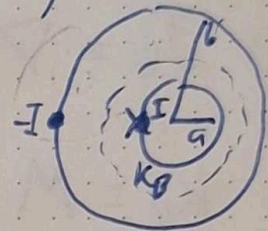
$$c) \mathbf{J}_b = \nabla \times \mathbf{M} = \frac{1}{s} \frac{\partial}{\partial s} \left(s \cdot \frac{I}{2\pi s} \chi_m \right) \hat{z} = 0$$

$$\mathbf{K}_b = \mathbf{M} \times \hat{n} = \frac{\chi_m I}{2\pi a} \hat{z} \text{ or } -\frac{\chi_m I}{2\pi b} \hat{z}$$

for $s=a$ & $s=b$.
Minus because I
goes opposite way
for $s=b$.

d) Total current for amperian loop $a < s < b$

$$\Rightarrow I + \mathbf{K}_b \cdot 2\pi a = (1 + \chi_m) I = I_{\text{enc}}$$

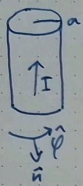


$$\Rightarrow \oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}$$

$$\Rightarrow 2\pi s |\mathbf{B}| = \mu_0 (1 + \chi_m) I$$

$$\Rightarrow \mathbf{B} = \frac{\mu_0 (1 + \chi_m) I}{2\pi s} \hat{\varphi}$$

4) a) $\uparrow +z$



$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{enc} \quad s < a$$

$$\Rightarrow 2\pi s |\mathbf{H}| = \frac{\pi s^2}{\pi a^2} I$$

$$\Rightarrow \mathbf{H} = \frac{Is}{2\pi a^2} \hat{\varphi}$$

$$\Rightarrow \mathbf{B} = \mu_0(1 + \chi_m) \mathbf{H} = \frac{\mu_0(1 + \chi_m) Is}{2\pi a^2} \hat{\varphi}$$

$s > a$

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{enc} \Rightarrow 2\pi s |\mathbf{H}| = I \Rightarrow \mathbf{H} = \frac{I}{2\pi s} \hat{\varphi}$$

$$\Rightarrow \mathbf{B} = \mu_0(1 + \chi_m) \mathbf{H} = \frac{\mu_0 I}{2\pi s} \hat{\varphi}$$

b) $\mathbf{M} = \chi_m \mathbf{H} = \frac{\chi_m Is}{2\pi a^2} \hat{\varphi}$

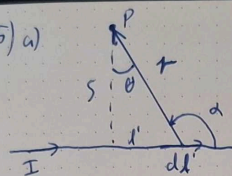
c) $\mathbf{J}_b = \nabla \times \mathbf{M} = \frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{\chi_m Is}{2\pi a^2} \right) \hat{z} = \frac{\chi_m I}{\pi a^2} \hat{z}$

$$\mathbf{K}_b = \mathbf{M} \times \hat{n} = -\frac{\chi_m I}{2\pi a} \hat{z}$$

$s=a$ at surface
right hand rule
gives $s - \hat{z}$

d) $\mathbf{I}_b = \mathbf{J}_b \cdot \pi a^2 + \mathbf{K}_b \cdot 2\pi a = \chi_m I \hat{z} - \chi_m I \hat{z} = \underline{\underline{0}}$

5) a)



Go through this in detail before, quick summary:

$$d\mathbf{l} \times \mathbf{r} = dl' \cos \theta$$

$$\sin \theta = \frac{dl'}{r} \quad \& \quad \cos \theta = \frac{s}{r}$$

$$\Rightarrow \tan \theta = \frac{dl'}{s} \Rightarrow dl' = s \tan \theta$$

$$dl' = \frac{s}{\cos \theta} d\theta \quad \& \quad \frac{1}{r^2} = \frac{\cos^2 \theta}{s^2}$$

$$\Rightarrow \mathbf{B} = \frac{\mu_0}{4\pi} I \int \frac{(d\mathbf{l}' \times \mathbf{r})}{r^2} = \frac{\mu_0}{4\pi} I \int_{\theta_1}^{\theta_2} \frac{\cos^2 \theta}{s^2} \left(\frac{s}{\cos \theta} \right) \cos \theta d\theta$$

$$\Rightarrow \mathbf{B} = \frac{\mu_0}{4\pi s} I (\sin \theta_2 - \sin \theta_1)$$

$$\Rightarrow B(s) = \frac{\mu_0 I}{2\pi s}$$

For infinite wire, $\theta_2 = \pi/2$ & $\theta_1 = -\pi/2$

b) The magnetic field at the second wire due to the first will be:

$$\mathbf{B} = \frac{\mu_0 I_1}{2\pi d}$$

c) The Lorentz force law gives

$$\mathbf{F} = I_2 \int d\mathbf{l} \times \mathbf{B} = I_2 B \int dl = \frac{\mu_0}{2\pi d} (3I_1 I_2) \int dl$$

and the force per length is then

$$\underline{\underline{F/L = \frac{3\mu_0}{2\pi d} I^2}}$$